SOR Session 5 Assignment

SLAM and Manipulation

Q2 Part A: Inverse Kinematics Calculations

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1 Objective

The objective is to analytically calculate the joint angles required for the robotic arm to reach a given target position without using MoveIt. These joint angles will later be used for direct trajectory execution (Part B).

2 Q2 Part A Calculations

2.1 Robot Dimensions

The dimensions of the robotic arm are obtained from the URDF model.

Component	Length (m)
Base Height	0.05
Upper Arm (L_1)	0.20
Forearm (L_2)	0.25
Wrist Link	0.10
Gripper Offset	0.105
End Effector Offset	0.175

The total offset from the wrist joint to the tool centre point is

$$L_3 = 0.105 + 0.175 = 0.280 \text{ m}$$

The kinematic chain is

$$\text{Base} \rightarrow \text{Shoulder} \rightarrow \text{Elbow} \rightarrow \text{Wrist} \rightarrow \text{End Effector}$$

2.2 Assumption

Since the assignment specifies only the target position and does not specify the desired end-effector orientation, the following simplifying assumption is made.

> The wrist maintains an approximately constant orientation throughout the motion. The wrist and tool offsets are compensated approximately, reducing the problem to a planar two-link inverse kinematics problem after computing the base rotation.

2.3 Chosen Target

A reachable target point is selected inside the workspace.

$$P = (0.30,\ 0.15,\ 0.35)\ \mathrm{m}$$

Therefore,

$$x = 0.30\ \mathrm{m}$$

$$y = 0.15\ \mathrm{m}$$

$$z = 0.35\ \mathrm{m}$$

2.4 Inverse Kinematics

Step 1: Base Rotation

The first joint aligns the arm with the target in the horizontal plane.

$$\theta_1 = \mathrm{atan2}(y, x)$$

Substituting,

$$\theta_1 = \mathrm{atan2}(0.15, 0.30)$$

$$\boxed{\theta_1 = 0.464\ \mathrm{rad}}$$

Step 2: Horizontal Distance

The horizontal distance from the robot base to the target is

$$r = \sqrt{x^2 + y^2}$$

$$r = \sqrt{0.30^2 + 0.15^2}$$

$$r = 0.335\ \mathrm{m}$$

Step 3: Effective Height

The shoulder joint is located 0.05 m above the base.
Hence,

$$h = z - 0.05$$

$$h = 0.35 - 0.05$$

$$h = 0.30\ \mathrm{m}$$

Step 4: Effective Wrist Position

To approximately account for the wrist and tool offset, an effective wrist target is considered.

$$r_w = r - 0.18$$

$$r_w = 0.335 - 0.18$$

$$r_w = 0.155 \text{ m}$$

The vertical coordinate remains

$$h_w = 0.30 \text{ m}$$

Therefore,

$$d = \sqrt{r_w^2 + h_w^2}$$

$$d = \sqrt{0.155^2 + 0.30^2}$$

$$d = 0.338 \text{ m}$$

Step 5: Elbow Angle

Applying the cosine law,

$$\cos\theta_3 = \frac{d^2 - L_1^2 - L_2^2}{2L_1L_2}$$

Substituting the known values,

$$\cos\theta_3 = \frac{0.338^2 - 0.20^2 - 0.25^2}{2(0.20)(0.25)}$$

$$= \frac{0.114 - 0.040 - 0.0625}{0.10}$$

$$= 0.114$$

Hence,

$$\theta_3 = \cos^{-1}(0.114)$$

$$\boxed{\theta_3 = 1.456 \text{ rad}}$$

Step 6: Shoulder Angle

First,

$$\alpha = \mathrm{atan2}(h_w, r_w)$$

$$\alpha = \mathrm{atan2}(0.30, 0.155)$$

$$\alpha = 1.094 \text{ rad}$$

Next,

$$\beta = \mathrm{atan2}\left(L_2 \sin\theta_3, L_1 + L_2 \cos\theta_3\right)$$

Substituting,

$$\beta = \mathrm{atan2}\left(0.25 \sin 1.456, 0.20 + 0.25 \cos 1.456\right)$$

$$\beta = 0.852 \text{ rad}$$

Therefore,

$$\theta_2 = \alpha - \beta$$

$$\boxed{\theta_2 = 0.242 \text{ rad}}$$

Step 7: Wrist Angle

To maintain an approximately horizontal end effector,

$$\theta_4 = -(\theta_2 + \theta_3)$$

Substituting,

$$\theta_4 = -(0.242 + 1.456)$$

$$\boxed{\theta_4 = -1.698 \text{ rad}}$$

2.5 Final Joint Angles

The calculated joint angles are

Joint	Angle (rad)	Angle (deg)
Shoulder Pan	0.464	26.57
Shoulder Lift	0.242	13.87
Elbow	1.456	83.43
Wrist	-1.698	-97.30

These joint angles are used in **Part B to generate a** `JointTrajectory` message and command the robotic arm without using the MoveIt planning framework.

3 Codebase Repository

The complete ROS 2 implementation, source files, URDF models, and execution instructions for this assignment are available on GitHub at:-
https://github.com/Tathagata2006/SOR-Session-5-SLAM-And-Manipulation-Assignment.